

2 (a) Define electric field at a point.

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..... [1]

(b) An oil droplet has charge of magnitude Q . The droplet is held in equilibrium by the electric field between two horizontal parallel plates, as shown in Figure 2.1.

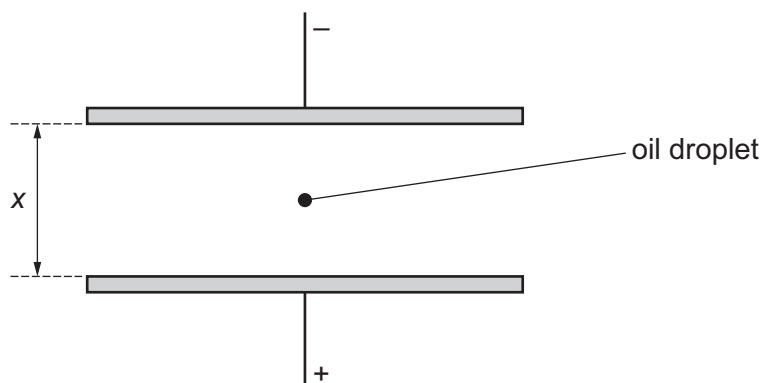


Figure 2.1

The electric field is created by a potential difference (p.d.) V between the plates, with the lower plate at the higher electric potential. The separation of the plates is x .

- (i) On Figure 2.1, draw four field lines to represent the electric field between the plates. [3]
- (ii) Explain, with reference to forces, whether the sign of Q is positive or negative.

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..... [2]

(c) For the oil droplet in **2(b)**, state expressions, in terms of some or all of Q , V and x , for:

(i) the electric field strength E

$$E = \dots\dots\dots [1]$$

(ii) the weight W of the droplet.

$$W = \dots\dots\dots [1]$$

(d) The plates in **2(b)** have a p.d. across them of 4800 V and a separation of 0.024 m. The magnitude of the charge on the oil droplet is $3e$, where e is the elementary charge.

Calculate the mass of the oil droplet.

$$\text{mass} = \dots\dots\dots \text{ kg } [2]$$

(e) The polarity of the p.d. across the plates in **2(b)** is reversed.

Describe and explain the subsequent motion of the oil droplet.

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 [2]

[Total: 12]